

Derivation of Roe Flux for Euler Equations and Isentropic Gas System

Euler Equations

The 1D Euler equations for inviscid compressible flow are:

$$\mathbf{u} = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u E + pu \end{bmatrix}$$

where ρ is density, u is velocity, E is total energy per unit mass, p is pressure, and $E = e + \frac{u^2}{2}$ with $e = \frac{p}{(\gamma-1)\rho}$ for an ideal gas (γ is the adiabatic index).

Roe Flux Process

1. **Jacobian:** The flux Jacobian is:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{1}{2}(\gamma-3)u^2 & (3-\gamma)u & \gamma-1 \\ -\frac{\gamma u E}{\gamma-1} + \frac{\gamma-1}{2}u^3 & \frac{\gamma E}{\gamma-1} - \frac{3-\gamma}{2}u^2 & \gamma u \end{bmatrix}$$

Eigenvalues: $\lambda_1 = u - c$, $\lambda_2 = u$, $\lambda_3 = u + c$, where $c = \sqrt{\gamma p / \rho}$. Eigenvectors:

$$\mathbf{r}_1 = \begin{bmatrix} 1 \\ u - c \\ H - uc \end{bmatrix}, \quad \mathbf{r}_2 = \begin{bmatrix} 1 \\ u \\ \frac{u^2}{2} \end{bmatrix}, \quad \mathbf{r}_3 = \begin{bmatrix} 1 \\ u + c \\ H + uc \end{bmatrix}$$

where $H = E + \frac{p}{\rho} = \frac{c^2}{\gamma-1} + \frac{u^2}{2}$.

2. **Roe Averaging:** For left ($\mathbf{u}_L = [\rho_L, \rho_L u_L, \rho_L E_L]^T$) and right ($\mathbf{u}_R$) states:

$$\tilde{\rho} = \sqrt{\rho_L \rho_R}, \quad \tilde{u} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}, \quad \tilde{H} = \frac{\sqrt{\rho_L} H_L + \sqrt{\rho_R} H_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}$$

Roe-averaged sound speed: $\tilde{c} = \sqrt{(\gamma-1) \left(\tilde{H} - \frac{\tilde{u}^2}{2} \right)}$. Eigenvalues: $\tilde{\lambda}_1 = \tilde{u} - \tilde{c}$, $\tilde{\lambda}_2 = \tilde{u}$, $\tilde{\lambda}_3 = \tilde{u} + \tilde{c}$.

3. **Wave Strengths:** Solve $\Delta \mathbf{u} = \mathbf{u}_R - \mathbf{u}_L = \sum_{k=1}^3 \tilde{\alpha}_k \tilde{\mathbf{r}}_k$:

$$\tilde{\alpha}_1 = \frac{\Delta p + \tilde{\rho} \tilde{c} \Delta u}{2 \tilde{c}^2}, \quad \tilde{\alpha}_2 = \Delta \rho - \frac{\Delta p}{\tilde{c}^2}, \quad \tilde{\alpha}_3 = \frac{\Delta p - \tilde{\rho} \tilde{c} \Delta u}{2 \tilde{c}^2}$$

where $\Delta \rho = \rho_R - \rho_L$, $\Delta u = u_R - u_L$, $\Delta p = p_R - p_L$.

4. **Roe Flux:**

$$\mathbf{F}_{i+1/2} = \frac{1}{2} (\mathbf{f}(\mathbf{u}_L) + \mathbf{f}(\mathbf{u}_R)) - \frac{1}{2} \sum_{k=1}^3 |\tilde{\lambda}_k| \tilde{\alpha}_k \tilde{\mathbf{r}}_k$$

Isentropic Gas System

For isentropic flow ($p = K\rho^\gamma$), the equations are:

$$\mathbf{u} = \begin{bmatrix} \rho \\ \rho u \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \rho u \\ \rho u^2 + \frac{K\gamma}{\gamma-1} \rho^{\gamma-1} \end{bmatrix}$$

Roe Flux Process

1. Jacobian:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -u^2 + K\gamma\rho^{\gamma-1} & 2u \end{bmatrix}$$

Eigenvalues: $\lambda_1 = u - c$, $\lambda_2 = u + c$, where $c = \sqrt{K\gamma\rho^{\gamma-1}}$. Eigenvectors:

$$\mathbf{r}_1 = \begin{bmatrix} 1 \\ u - c \end{bmatrix}, \quad \mathbf{r}_2 = \begin{bmatrix} 1 \\ u + c \end{bmatrix}$$

2. Roe Averaging:

$$\tilde{\rho} = \sqrt{\rho_L \rho_R}, \quad \tilde{u} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}, \quad \tilde{c} = \sqrt{K\gamma \tilde{\rho}^{\gamma-1}}$$

Eigenvalues: $\tilde{\lambda}_1 = \tilde{u} - \tilde{c}$, $\tilde{\lambda}_2 = \tilde{u} + \tilde{c}$.

3. Wave Strengths: Solve $\Delta \mathbf{u} = \tilde{\alpha}_1 \tilde{\mathbf{r}}_1 + \tilde{\alpha}_2 \tilde{\mathbf{r}}_2$:

$$\tilde{\alpha}_1 = \frac{\Delta(\rho u) - \tilde{u}\Delta\rho + \tilde{c}\Delta\rho}{2\tilde{c}}, \quad \tilde{\alpha}_2 = \frac{\Delta(\rho u) - \tilde{u}\Delta\rho - \tilde{c}\Delta\rho}{-2\tilde{c}}$$

4. Roe Flux:

$$\mathbf{F}_{i+1/2} = \frac{1}{2} (\mathbf{f}(\mathbf{u}_L) + \mathbf{f}(\mathbf{u}_R)) - \frac{1}{2} \sum_{k=1}^2 |\tilde{\lambda}_k| \tilde{\alpha}_k \tilde{\mathbf{r}}_k$$

Both derivations ensure a conservative, consistent flux satisfying the Rankine-Hugoniot condition.